EQUIPMENT NAME:	**FIM III Booster Station Switchroom**
EQUIPMENT NUMBER:	**F23-SR-601**

Procurement Package	"EL006" - Switchroom
Equipment Number	**Equipment Name**
F23-SR-601	FIM III Booster Station HV/LV Switchroom
Major Equipment in Switchroom	
Equipment Number	**Equipment Name**
F23-TR-602	FIM III Booster Station 11/0.433kV Transformer (refer to datasheet 2000-F23-DSH-EL-40002)
F23-SW-601	FIM III Booster Station HV Switchboard (refer to datasheet 2000-F23-DSH-EL-40004)
F23-SW-602	FIM III Booster Station LV Motor Control Centre (refer to datasheet 2000-F23-DSH-EL-40005)
F23PP651-VS	FIM III Booster Pump 1 Variable Speed Drive (refer to datasheet 2000-F23-DSH-EL-40006)
F23PP652-VS	FIM III Booster Pump 2 Variable Speed Drive (refer to datasheet 2000-F23-DSH-EL-40006)

Reference Documents		
Document Number	**Rev**	**Description**
Datasheets		
2000-F23-DSH-EL-40002	0	FIM III Booster Station 11/0.433kV Transformer
2000-F23-DSH-EL-40004	0	FIM III Booster Station HV Switchboard
2000-F23-DSH-EL-40005	0	FIM III Booster Station LV Motor Control Centre
2000-F23-DSH-EL-40006	0	FIM III Booster Pump Variable Speed Drives
Equipment Lists		
2000-F23-LST-EL-40001	0	FIM III Booster Station Electrical Load List
2000-F23-LST-EL-40002	0	FIM III Booster Station Electrical Equipment List
2000-F23-LST-EL-40003	0	FIM III Booster Station Cable Schedule
2000-F23-LST-EL-40006	0	FIM III Booster Station LV Components List
Drawings		
2000-F23-BLK-IC-40001	A	F23-SR-601 Switchroom Communications Block Diagram
2000-F23-DRG-EL-40002	0	F23-SW-601 HV Switchboard Remote Switching Panel General Arrangement
2000-F23-DRG-EL-40011	1	F23-DB-601 Switchroom and Area L&SP DB General Arrangement
2000-F23-DRG-EL-40012	1	F23-DB-602 Switchroom HVAC DB General Arrangement
2000-F23-DRG-IC-40001	0	F23-COM-601 Communications Cabinet General Arrangement
2000-F23-DRG-IC-40002	0	F23-DCS-601 DCS Cabinet General Arrangement
2000-F23-DRG-IC-40003	0	F23-RTU-606 RTU Cabinet General Arrangement
2000-F23-SCM-EL-40001	0	F23-SW-601 HV Switchboard Remote Switching Panel Schematic Diagram Sheet 1
2000-F23-SCM-EL-40002	0	F23-SW-601 HV Switchboard Remote Switching Panel Schematic Diagram Sheet 2
2000-F23-SCM-EL-40003	0	F23-SW-601 HV Switchboard Remote Switching Panel Schematic Diagram Sheet 3
2000-F23-TER-EL-40001	0	F23-SW-601 HV Switchboard Remote Switching Panel Termination Diagram
2000-F23-TLD-EL-40001	0	F23-BC-601 Switchroom 48VDC Battery Charger Three-Line Diagram
2000-F23-TLD-EL-40011	0	F23-DB-601 Switchroom and Area L&SP DB Three-Line Diagram
2000-F23-TLD-EL-40012	0	F23-DB-602 Switchroom HVAC DB Three-Line Diagram
2000-F23-TLD-EL-40013	0	F23-DB-603 Switchroom 240V AC UPS DB Three-Line Diagram

Reference Specifications
2000-F00-STS-EL-10004_2 – High Voltage Electrical Testing Specification
2000-F00-STS-EL-10007_0 – Low Voltage Distribution Board Specification
2000-F00-STS-EL-10008_0 – Electrical Enclosures Specification
2000-F00-STS-EL-10009_0 – Cables Specification
2000-F00-STS-EL-10011_0 – Switchrooms Specification
2000-F00-STS-EL-10014_2 – Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification
2000-F00-STS-EL-10015_0 – High Voltage Switchboards & Switchgear Specification
2000-F00-STS-EL-10016_1 – Low Voltage Switchboards & Switchgear Electrical Specification
2000-F00-STS-GE-10001_2 – Packing and Shipping Specification
2000-F00-STS-GE-10002_0 – Site Specific Information
2000-F00-STS-GE-10003_2 – Plant Numbering Specification
2000-F00-STS-ST-10001_2 – Steelwork Fabrication
2000-F00-STS-ST-10002_2 – Steelwork Erection Specification
SPEC-00-G-05_12 – Technical Specification for Protective Coatings

Standards
AS 1170 – Structural Design Actions
AS 1670 - Fire detection, warning, control and intercom systems - System design, installation and commissioning
AS 2067 – Substations and High Voltage Installations Exceeding 1kV AC

AS 2293 – Emergency Escape Lighting and Exit Signs for Buildings
AS 2676 – Guide to the Installation, Maintenance, Testing and Replacement of Secondary Batteries in Buildings
AS 3000 – Electrical Installations (known as the Australian Wiring Rules)
AS 4100 – Steel Structures

Switchroom Description

1. 11kV and 415V transportable type switchroom comprising:

- Transportable building – standalone structure mounted on a structural steel base frame with supporting columns, landings (at equipment entry and personnel entry ends) and stairs;
- Switchroom building will be comprised of the following (refer to switchroom layout drawing);
  - HV switchroom
  - Airlock
  - LV switchroom
  - Transformer room
- Airlock shall house Remote Switching Panel(s) for safe operation of 11kV switchgear (all panels)
- Airlock shall house NHP TemRack IR Local Control Station(s) for safe operation of LV air circuit breaker incomer.
- Shrouded Emergency Shutdown Pushbutton as noted on the drawing shall be located inside the airlock. Emergency Shutdown shall trip all incomers and upstream feeder circuit breakers (Field wiring by others).
- Fit for purpose air-conditioning – preferably industrial split systems, appropriate for the heat loads with N+1 redundancy. The redundant unit shall maintain switchroom temperatures within limits when one unit fails. Air-conditioning shall be tripped on fire alarm.
- Fire detection and suppression systems comprising:
  - Smoke and heat detectors
  - Manual call points
  - VESDA system
  - Fire Indicating Panel
  - Inergen (or equivalent) gas suppression system inclusive all pipe works. Gas canisters shall be located indoors
  - CO_2 fire extinguishers inside and outside as per National Construction Code (NCC) Requirements including signage
  - All signages.
  - VESDA system shall sample air from the following:
    - Switchroom ceiling – HV switchroom, LV switchroom and Transformer Room
    - HV switchgear (LV compartment only)
    - Incomer and cable zones of each tier in LV MCCs
    - LV VSDs > 315kW
    - HV VSDs
    - Communications cabinet
    - DCS and RTU panel
- Switchroom LV DBs, lighting (including emergency lights), GPOs and RJ45 outlets, defibrillator, rescue kits and CPR signage
- All doors leading from the airlock into switchrooms shall have a 200 x 200mm laminated safety glass window to allow operators to see inside the room for fire/smoke prior to entry.
- Panic bars shall be provided on all egress doors.

2. The Switchroom shall house the following electrical equipment as per the electrical equipment lists and as follows:

- 11kV and 415V switchgear, dry type transformer, HV and LV VSDs, 240V AC UPS & batteries, 48V DC battery charger and batteries, distribution boards, fire panels, HVAC equipment, DCS panel, RTU panel, switchroom IO and communications cabinet
- 11/0.433kV dry type transformer (IP00) shall be housed inside a purpose-built transformer room with air conditioning (N+1)
- Transformer room shall have a removable wall panel (or double doors) for transformer change out in case of failure. If a removable panel is used it shall be fitted with a personnel access door. All access door(s) shall be equipped with a Fortress key lock to ensure supply power is isolated and earthed prior to personnel entry - Fortress keys to be fitted on supply feeder earth switch and keyed alike to Transformer room personnel access doors
- Transformer installation and removal into and out of the room shall be facilitated with C-channels installed into/onto the transformer enclosure floor. The dimensions and spacing of the C-channels shall be determined by the wheel and spacing dimensions of the dry transformers. C-channels shall be extended onto the external platform allowing the transformer to be pushed into the room or pulled out of the room if required.

3. Special requirements

- Vendor shall supply, install and test protection relays:
  - SEL Protection Relays (as per tables below and Single Line Diagrams) with common functions:
    - SEL 751 protection relays with integral arc flash (where available), SEF protection and Modbus TCP and IEC 61850 communications and dual RJ45 ethernet ports
    - SEL 787 Transformer differential protection relay
    - SEL710 motor protection relays
  - SEL735 Power Meters with Modbus TCP communications
  - Supplementary SEL 751#JP0G used as arc flash detection relays where SEL relays do not have native arc flash functionality (ie. on 11kV Incomers with SEL787 relays)
  - Common CT specifications:
    - SE-CS30 current transformer for NER
  - Common VT specifications: 11/√3 / 0.11/√3 kV 15VA CL0.2 & 20VA CL 3P.
- LV MCC arc fault detection current transformers to be installed in the transformer room on purposed build frame on the transformer's secondary.

Scope of Work

The vendor shall design, fabricate, assemble, test and transport (to site) the transportable switchroom, including installation, wiring, terminating, and testing as far as reasonably practical, all equipment listed as supplied and/ or installed by "EL006" in the equipment lists (including all free-issued equipment), prior to the switchroom being shipped to site.

Work Included

- Design, supply, fabrication, fit-out, and testing of one (1) transportable switchroom building, including but not limited to:
  - Paint system CS1 (or equivalent), structurally certified base frames, stairs;
  - Paint system CS1 (or equivalent), structurally certified, equipment platforms (rated for a minimum of 10kPa) and switchroom support columns;
  - Switchroom building;
  - Design, supply and installation of one Remote Switching Panel per switchboard to be mounted inside the airlock for switching/control of all 11kV circuit breakers and fused contactors, complete with switches and indication semaphores (based on the Single Line Diagrams);
  - Supply and installation of one NHP TemRack IR LCS per LV MCC Incomer;
  - Normal, emergency and external lighting;
  - General power and network (RJ45) outlets;
  - Design, supply and installation of air conditioning system based on switchroom heatload, for each room. Vendor to supply copy of calculations to Principal for review.
  - Air conditioning, complete and operational (to be tested during switchroom FAT) with piping, drains and isolators, to be installed such that it can remain attached during transport (if practical);
  - Air conditioner controller, one per room (Including transformer enclosure), with integral temperature sensing. To be wired directly to the air conditioner units.
  - Room combined temperature and humidity transmitter, one per room. Hardwired analogue output to DCS;
  - Lighting & small power and services 415V distribution board;
  - HVAC 415V distribution board;
  - 240V AC UPS, battery cabinet and batteries, and distribution board. Note that batteries shall be delivered separately to site.;
  - Communications cabinet and fit out in accordance with Switchroom equipment and cable lists;
  - Active Harmonic Filters as specified, including inter-cabling and CTs;
  - Fire detection equipment, fire indication panel and VESDA;
  - Gas suppression systems including Inergen (or equivalent) gas cylinder/s, valves, piping and cabinets;
  - 415V MCC;
  - Incomer ACB Arc Flash CTs to be mounted on LV side of the 11 / 0.433 kV transformer;
  - Installation of free-issue equipment (refer to equipment list), including:
    - 11 / 0.433 kV dry type transformer
    - HV Switchboard
    - HV VSDs
  - Installation of free-issued HV Switchgear Core balance CTs – to be mounted by Vendor in the drop box of the associated switchgear tier.
  - 48VDC battery charger and batteries;
  - Allowance for any ducting and penetrations as required by equipment OEMs;
  - Conduit, cable tray and cabling between all equipment installed in or supplied with the room; or inter wiring between equipment located in combined switchroom F23-SR-601.
  - HV, LV, communications and instrument cabling listed as supplied and/ or installed by "EL006" in the Cable Schedule
  - Supply, installation and termination of switchroom cabling, whether listed or not in the cable schedule, for cabling and connections between switchroom equipment, including but not limited to:
    - Fire systems
    - Internal communications cabinet cabling
    - Internal RTU and DCS panel cabling
    - Internal LV MCC cabling
    - HVAC cabling
    - Lighting and small power
    - UPS and battery chargers
  - Engagement of equipment OEM for support for equipment installation and testing. Specific engagement shall be sought with ABB for HV Switchgear CT installation and function testing of interlock systems;
  - Earthing of equipment in the room to the switchroom earth bar;
  - Equipment nameplates and labels;
  - Cable labels;
  - Signages;
  - Switchroom landings, support structure and stairs (except for above stated exceptions);
  - Installation of key exchange boxes where specified in the SLD's;
  - Any other items which are normally supplied with this equipment.
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- Factory Acceptance Testing at manufacturer's premises including:
  - FAT assistance for Client or Client's Representative FAT and construction inspection – provide temporary power and electrician for 5 days to assist with FAT
  - CT / relay injection testing
  - Gas suppression system pressure testing
- Supply of all necessary documentation, calculations where applicable, drawings and test documentation;
- Suitable internal bracing for road transport to site Owner FIM
- Spare parts list to be provided;
- Unloading and storage of free-issued equipment. Acceptance of free-issued equipment from September 2024;
- Transportation of the switchroom, HVAC units (if separate), landings & stairways in compliance with Main Roads regulations to the delivery point.

Work Excluded

- Fabrication / supply or design of free-issue items;
- Offloading the equipment at site;
- Installation and commissioning beyond support specified above.

Description	Unit	Duty	Vendor	Notes	Rev
General					
Type of Building		Transportable Switchroom		1	0
Method of Installation		Platform mounted, Bottom Entry		1	0
Codes		Building Code of Australia (BCA) and FESA		1	0
Compliance with Applicable Australian Standards		Noted above		1	0
Equipment Layout Drawing		Refer to GA – drawing is indicative only, vendor to provide construction layout for review/ approval	Offer is based on vendor drawing T5950-SL-001_A	3, 4	0
External Layout Drawing		By vendor	To be provided at detail design stage	3, 4	0
Penetrations Drawing		By vendor, allow for drop boxes and penetrations below HV switchboards, 415V MCCs, HV and LVVSDs, UPS, DBs, Comms and other cabinets		1, 4	0
Manufacturer			POWINS	2	0
Country of Manufacture			Australia	2	0
Environmental Conditions					
Plant Location		Western Australia		1	0
Service Location	Indoor/ Outdoor	Outdoor, direct sunlight and rain		1	0
Site Data		2000-F00-STS-GE-10002 - Site Specific Information		1	0
Structural Design Criteria					
Wind Loading		AS 1170.2		1	0
Earthquake Loading		AS 1170.4		1	0
Site Data		2000-F00-STS-GE-10002 - Site Specific Information		1	0
Support Reactions		To be determined in detailed design		2, 6	0
Construction					
Building Steel Base		Rigid self-supporting base frame with certified lifting facilities, Subfloor cable ladders to be located within baseframe.		1,7	0
Lifting Lugs		Yes – stamped with safe working load		1	0
Lifting		Switchroom shall be provided with lifting diagram and shall be sufficiently rigid to lift with multiple spreader bars	Lifting Diagram is provided for the lifting contractor to conduct a lifting study. However the rooms are designed for multiple point lift down each side of the room with the use of more than 1 spreader bar.	3	0
Building Material for Walls and Roof		Welded steel stud and structural portal frames, insulation in cavities, with color bond internal (shallow profile) and external cladding (trim deck profile) COLORBOND® Steel - TRIMDEK® or approved equivalent	POWINS bolted construction method only	3	0
Materials & Thickness of Insulation		Vendor to meet temperature requirements, minimum as below:		1	0
Wall	mm	Minimum R3.0 Rockwool Insulation Batts	Yes	3	0
Ceiling	mm	Minimum R3.0 Rockwool Insulation Batts	Yes	3	0
Moisture Barrier		Sarking foil on portal frame	Sarking foil on portal frame	3	0
Roof Slope Orientation		Skillion 3 degrees	Skillion 5 degrees	3	0
Roof Gutters and Down Pipes		No		1	0
Flooring		Compressed fibre cement (minimum thickness of 18mm) with non-flammable industrial grey vinyl (minimum thickness of 2mm) with black skirting		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Windows		Access/exit doors to be equipped with a minimum 200mm by 200mm laminated safety glass window.		1	0
Air Lock Room		Yes, air lock entry for personnel access door.		1	0
Egress doors		All egress doors to be equipped with panic bar.		1	0
Switchroom doors – Personnel Access to Airlock		Access door(s) with laminated 200x200mm safety glass window for each room leading off airlock.		1	0
Switchroom doors - Equipment Access		One two-leaf equipment door onto equipment deck. Door swing to both be 170 degrees with door catch to allow maximum deck space when maneuvering equipment on deck.		1	0
Transformer room – General		Structurally designed to suit the dry-type transformer weight (up to7000kg); transformer to be permanently fixed with wheels removed and securely stored within the transformer room.		1	0
Transformer room –Transformer Room access		One personnel access door WITH Fortress key lock – keyed alike to upstream HV circuit breaker earthswitch. Include removable wall panels and rails for transformer removal. One personnel egress door into the LV switchroom. Egress only, no access from switchroom.		1	0
Transformer room - Gland and kit plates		Gland plate and kick plate to be mounted on floor under both the HV and LV sides of the transformer.		1	0
Transformer room - C Channel		C-channel to be mounted in the floor such that the top of the channel sits flush with the floor.Channel shall be mounted the same width as shown on transformer GA.		1	0
Material for Internal Walls		Non-Combustible (Stramit Minirib)	Yes (Stramit Minirib)	3	0
Internal ceiling and wall finishing colour		Surfmist or equivalent	Surfmist (Stramit Minirib)	3	0
Internal floor colour		Grey preferred	Confirmed	3	0
External painting compliant with specification surface protection system CS1		Yes	Confirmed	3	0
Switchroom external colour		Lysaght Shale Grey		1	0
Base frame member surface protection system CS1		Shale Grey or equivalent	Confirmed	3	0
Dimension from bottom of subframe to finishedfloor level			520mm	2	0
Cable ladder (mounted under floor inside baseframe members)		Stainless Steel 316 Cable Ladder (600mm)		1	0
Support Structure					

Description	Unit	Duty	Vendor	Notes	Rev
Supply of support structure	Yes/No	Yes, building columns / legs shall be 1.8m high from the ground to the underside of the building steel base frame. Support structure to include cantilever supports for 2 laterally (side by side), 3 rows high (vertically), 600mm cable ladders to run on both sides under the switchroom for field wiring. Cable ladders will be supplied by others. Surface coating - paint system CS1 or equivalent	Confirmed	3	0
Supply of stairs	Yes/No	Yes		1	0
Supply of landings	Yes/No	Yes. Include C channels / rails embedded flush to landing where applicable outside of transformer room for install / removal purposes of transformer		1	0
Provided with cable ladder supports for subfloor	Yes/No	Yes, VTA details	Confirmed	2	0
Bolted for ease of transport and assembly on site	Yes/No	Yes		1	0
Landings to be attached to switchroom for site landing	Yes/No	Yes		1	0
Support structure/legs, landings and stairs surface coating		Paint system CS1 or equivalent		1	0
Support structure paint colour		Shale Grey or equivalent		1	0
External Cable Entry					
Cable Entry		Bottom Entry via Floor		1	0
Cable Drop Boxes Provided		Required under HV switchgear, 415V MCC, HV and LV floor mounted VSDs, AHFs and all floor mounted equipment. Constructed with removable side and bottom gland plates. To be within shipping envelope of building and remain installed during transport. Vendor to provide detail drawings of offer. Surface coating - paint system CS1 or equivalent	Side access drop boxes are offered	3	0
Floor Penetrations for Cable Entry		As above and locate to avoid floorbeams and joists, sealed. Where gland plate is exposed on finished floor surface, kick plate shall be installed. E.g. under wall mounted DBs		1	0
Floor Penetration Gland Plates		5mm aluminium, earthed with neoprene gasket		1	0
Mechanical Services Requirements					
Internal Room Temperature to be maintained	°C	21°C to 27°C		1	0
Fouling Factor to be applied	%	20		1	0
Airconditioning Design requirements		Air conditioners, complete with piping and isolator, as required to maintain internal ambient temperature and provide N+1 redundancy with 20% fouling factor		1	0
Split system make and model			Mitsubishi Electric	2	0

Description	Unit	Duty	Vendor	Notes	Rev
Air conditioner types - HV switchroom		Multiple Split Systems (N+1) *preferred* – otherwise, high capacity commercial ducted or CRAC systems installed on equipment platforms. (*Non-preferred are separate ground mounted units*).	Under ceiling, cassette type split systems as per POWINS offer T5950-250130-SW-Owner SWRM_C	3	0
Air conditioning types – All other (non-HV) rooms		Multiple Split Systems (N+1) – single phase, wall mount or 3 phase, in ceiling, high kW commercial type units with individual isolators	Under ceiling, cassette type split systems as per POWINS offer T5950-250130-SW-Owner SWRM_C	3	0
Air conditioner internal condensation drip trays (splits)		To be included under each unit (on wall or ceiling where relevant), unless there are built-in drip trays (e.g. ceiling cassettes)	Included where required	3	0
Air conditioner temperature control		Single wall mounted controller for each room, to cover all split systems within the room.		1	0
Air conditioner controller make/model			Mitsubishi PAR-40MMA	2	0
Air conditioner controller communication		Modbus TCP capable	No	3	0
Air conditioner controller status contacts to DCS		Yes, 2 relay outputs to DCS (Running, Fault) per room.	No	3	0
Air conditioner controller - FIP		FIP to control contactors for A/C fire system trip	Confirmed	1	0
Air conditioner cooling capacity, rating and quantity		Vendor to complete Table B below.	See table	2	0
Air conditioner fully installed at factory and tested		AC units to be fully installed at the factory (includes the outdoor unit, piping/electrical connections), tested during switchroom FAT, and to remain fully assembled for transport (preferred)		1	0
A/C outdoor unit installed for transport		Yes – outdoor units are installed on ends of Switchroom and transported attached to room. Vendor to advise for larger units		1	0
Combined Temperature & Humidity Sensor		Yes – one per switchroom and transformer room. Not required in airlock. 4-20mA output to DCS. Vendor to list make / model	Confirmed. Siemens QFA2071 Temperature / Humidity Transmitter	2	0
Pressurisation fan		No		1	0
LV switchgear heat load		Vendor to complete Table A below		2	0
VSD heat load		Vendor to complete Table A below		2	0
Switchroom internal equipment heat load		Vendor to complete Table A below		2, 11	0
Switchroom environmental heat load (Passive heatload)		Vendor to complete Table A below		2	0
Remote Switching Panels					
11kV switchboard	Yes/No	Yes		1	0
Control Voltage		48VDC		1	0
Indicator Light Type		Semaphore	Confirmed	3	0
LV MCC TemRack IR LCS	Yes/No	Yes, just the incomer		1	0
Control Voltage		24VDC		1	0
Indicator Light Type		Semaphore	Confirmed	3	0
Location		In airlock		1	0
Emergency Shutdown Pushbutton					
Design, Supply and Install	Yes/No	Yes, two off (2)		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Design		Red mushroom head latching push button comes with yellow ring, and a shroud or lift-up flap to prevent inadvertent operation. Clear warning signage required.		1	0
Contacts		2 N/C contacts per LV incomer 2 N/C contacts per HV incomer 1 N/C contact for DCS		1	0
Location		1. Airlock 2. Outside HV Room		1	0
LV MCC					
Floor Space for Future Expansion		Vendor shall provide floor space adequate for one future functional double-sided tier and cable zone tier on the extendable side of the MCC. Vendor to advise if steel work, floor cutout and drop boxes have been allowed for.	Confirmed. Room size based on vendor drawing T5950-SL-001_A. MCC length 4.92m	3	0
Arc Fault Detection – Current Transformer		Vendor to install current transformers on purposely built frame in the Transformer Room		1	0
Electrical Services					
Supply and Installation of the below electrical services:					
Interior Normal Lighting and Switches	Yes/No	Yes, internal lighting and light switches		1	0
Emergency and Exit Lighting to AS2293 suppliedand installed by vendor	Yes/No	Yes, emergency and exit lighting to AS2293		1	0
External Wall Mounted Luminaires	Yes/No	Yes, LED lights mounted above each egress doorway. One external wall mounted LED flood light on each side of the building. Wall mount emergency lighting toAS2293 required for all landing stairway.	External lights based on the quantity and locations shown on the tender GA drawing	3	0
Photoelectric cell	Yes/No	Yes		1	0
Small Power	Yes/No	Yes, internal GPOs. A/C isolators (one per A/C)		1	0
L&SP Distribution Board	Yes/No	Yes, 3 Phase L&SP DB to supply building services, with auto/manual selector for externallighting, photocell relay output tocontrol auto external lighting. Minimum 20% spare capacity, and Rapid Test RCD Test system included.		1	0
415V HVAC Distribution Board	Yes/No	Yes, Vendor to size DB to suit the required number of air conditioning Circuits.		1	0
240VAC UPS Distribution Board	Yes/No	Yes, 240VAC UPS Distribution Board for FIP, communications, MCC control supply, DCS and RTU supplies.		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Cabling for Mechanical, Electrical, Comms and Fire Services	Yes/No	Yes, cabling for mechanical, electrical and fire services. Cabling and supports between all room equipment (including free-issued equipment) as required. For free issue equipment cabling, refer to the Electrical Equipment List and Cable Schedule. Vendor is responsible for cable sizing of all cables between switchroom equipment.		1, 7, 8, 9	0
Communication Cable segregation		For all equipment mounted against the wall, communications cables are to be run in duct, mounted above the equipment, to the communications cabinet. For equipment mounted in the centre of the room, there shall bea dedicated, segregated, 150mm cable ladder, mounted in the switchroom subframe, to the communications cabinet.		1	0
Wall mount VSD Support	Yes/No	Yes, Vendor to design and install a purposedly built wall frame to support the wall mounted LV VSDs including spare space for future use		1	0
Main Earth Bar	Yes/No	Yes, pre-drilled and sized to suit all switchroom internal connections,plus 12 external connections and 20% spare capacity		1	0
Earthing Cables	Yes/No	Yes, including earthing of free-issued and supplied equipment and building structure.		1, 7	0
Fire Services and Safety					
Fire panel, manual buttons and sensors supplied and installed as shown on drawings	Yes/No	Yes		1	0
Fire Panel make		Wormald	Tyco	3	0
Fire Panel model		Vigilant 8 Zone F3200	MX1 Addressable Fire Panel	3	0
Design notes		N/B		1	0
Smoke and heat detector make and model		Photo Optical	TBC during detailed design	3	0
Aspirating Smoke Detector (ASD) make and model		Xtralis VESDA VEP-A00-P	Confirmed	3	0
Aspirating Smoke Detector (ASD) sample points		Multiple pipes with LED aspirating system. Multiple sampling points, transformer enclosures, aspirating tube along ceilings, LV compartments of HV switchgear, LV MCC cable zones, VSDs >315kW, DBs >100Amps, DCS cabinets, RTU panels, UPS cabinet. Quantity TBC by vendor during detailed design	Confirmed	3	0
Manual glass break type alarm buttons make and model		Gas local control call point	Confirmed	3	0
Fire panel input voltage		240VAC 50Hz single phase from UPS DB		1	0
Backup battery	Yes/No	Yes		1	0
Fire Panel Communications		**Refer 2000-F23-BLK-IC-40001**		1	0
Spare relay contact (for status/alarm to DCS)	Yes/No	Yes – Alarm, Isolate, Fault		1	0
Automatic trip to air-conditioners	Yes/No	Yes		1	0
Gas suppression systems – General	Yes/No	Yes, incl. Inergen (or equivalent) gas cylinder/s, valves, piping and cabinets; Cylinder/s to be installed indoor within the Switchroom.		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Gas suppression systems - Cylinder Gas type		IG541 (Frike Pro Inert)	Confirmed	3	0
Gas suppression systems - Cylinder Qty		As commercial offer	5 x 300bar 80L	3	0
Gas suppression systems - Cylinder Weight and dimension (Each)		As commercial offer	TBC during detailed design	3	0
Fire Extinguisher type		5kg CO_2 AS PER AS 2444		1	0
Fire extinguisher wall mount brackets and sign provided	Yes/No	Yes		1	0
Fire blankets provided	Yes/No	Yes, 2 off 1.2 x 1.8m		1	0
Fire blankets mounted on walls	Yes/No	Yes		1	0
Fire aid kit and LV resuscitation kit supplied and installed as shown on drawings	Yes/No	Yes		1	0
Safety signage and resuscitation chart supplied and installed	Yes/No	Yes		1	0
HV Rescue Kit provided	Yes/No	Yes – in each HV switchroom		1	0
LV Rescue Kit provided	Yes/No	Yes		1	0
Automatic external defibrillator		Lifepak Defibrillator CR2 Fully Automatic	Confirmed	3	0
Equipment within Switchroom					
Supply and Installation of Equipment and Cabling		As per the equipment list and cable schedule		1	0
Labelling					
Switchroom Entrance Label					
Access Doors		Yes, switchroom identification label, danger sign, voltage warning sign		1	0
Transformer Room Access – Additional Label		Yes, isolate before entry label, high voltage warning sign		1	0
Arc Flash Labels	Yes/No	Yes, Engineer to supply label data, Vendor to supply and install labels. Labels to be fitted on all HV switchgear, HV VSDs, LV MCCs and >100A rated LV DB/VSDs		1	0
Traffolyte label required on switchroom equipment (including free issue equipment supplied without labels)	Yes/No	Yes, etched traffolyte label (blackon white background) on switchroom equipment.		1	0
Label Information		Equipment number, equipment description, and schematic drawing number.		1	0
“Danger xxxV” labelling on equipment suppliedwith voltages above ELV		Yes, traffolyte label. Traffolyte“Fed from” label with suppliedfrom equipment/circuit also required.		1	0
Emergency Shutdown Pushbutton Label		Labelled with” “EMERGENCY SWITCHROOM SHUTDOWN – ACTIVATING THIS PUSHBUTTON WILL TRIP THE SWITCHBOARD INCOMING POWER SUPPLY”		1	0
Lighting and Outlets		Labelled with circuit number.		1	0
Test Requirements					
Test Required					
Inspection		Yes		1	0
Routine		Yes		1	0
Special		Yes, including L&SP, Rapidtest system, Air conditioning system and Fire systems (inclusive of room air tightness integrity test)		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Commissioning		Full off-site commissioning of work completed by Vendor. Allowance for 5 days of integration testing to be performed by Engineer at Vendor's premises. Temporary power to be available to switchboard for the duration of the integration testing.		1	0
Shipping and Delivery					
Overall Dimensions of Building	L x W x H (mm)	Vendor to complete below **Table C** with switchroom dimensions. Length, Width and Height (for shipping and internal ceiling height		2	0
Total Weight of Building	kg	Vendor to complete below **Table C** with approximate shipping weights for tender. Includes building, attached platforms & installed equipment		2	0
All penetrations sealed before shipping	Yes/No	Yes		1	0
All loose items, such as external light fittings and fire extinguishers are packed into switchroom before shipping	Yes/No	Yes		1	0
All batteries to be demounted and packed into switchroom for transport.	Yes/No	Yes		1	0
Temporary bracing provided for any equipment requiring additional support before shipping provided	Yes/No	Yes		1	0

Description	Unit	Duty	Vendor	Notes	Rev
HV Switchboard Protection Relays					
Feeder Protection Relays					
Manufacturer		SEL	Included	3	0
Model		SEL-751#JP0G		3	0
Supply Voltage	V	48VDC		1	0
Communications Interface		As per SEL-751#JP0G		3	0
Communications Protocols Supported		As per SEL-751#JP0G		3	0
Protection Relay Features					
Voltage Sensing	Yes/No	As per SEL-751#JP0G	Included	3	0
Tripping Supply Monitoring	Yes/No			3	0
CT Saturation Detection	Yes/No			3	0
Harmonic Blocking (2nd & 5th)	Yes/No			3	0
Directional Protection	Yes/No			3	0
Undervoltage Protection	Yes/No			3	0
Underfrequency Protection	Yes/No			3	0
Transformer Protection Relays					
Manufacturer		SEL	Included	3	0
Model		SEL-787#22EK (07872EE2CCC0X78810690)		3	0
Supply Voltage	V	48VDC		1	0
Communications Interface		As per SEL-787#22EK Transformer Protection Relay		3	0
Communications Protocols Supported				3	0
Protection Relay Features (essential)					
2 Winding Differential	Yes/No	As per SEL-787#22EK Transformer Protection Relay		3	0
Tripping Supply Monitoring	Yes/No			3	0
CT Saturation Detection	Yes/No			3	0
Harmonic Blocking (2nd & 5th)	Yes/No			3	0
LV Restricted Earth Fault	Yes/No			3	0
Transformer Alarms &Trips					
Buchholz	Yes/No	As per SEL-787#22EK Transformer Protection Relay		3	0
Winding Temp.	Yes/No			3	0
Fluid Temp.	Yes/No			3	0
Pressure Relief	Yes/No			3	0
Power Quality Meters					
Manufacturer		SEL	Included	3	0
Model		SEL-751#L54R		3	0

Description	Unit	Duty	Vendor	Notes	Rev
Supply Voltage	V	48VDC		1	0
Communications Interface		As per SEL-751#L54R		3	0
Communications Protocols Supported				3	0
Generator Protection					
Manufacturer		SEL	Included	3	0
Model		SEL-751#JP0G (751002CCC0X70811F90)		3	0
Supply Voltage	V	48VDC		1	0
Communications Interface		As per SEL-751#JP0G		3	0
Communications Protocols Supported				3	0
Motor Protection					
Manufacturer		SEL	Included	3	0
Model		SEL-751#55RX		3	0
Supply Voltage	V	48VDC		1	0
Communications Interface		As per SEL-751#55RX Motor Protection Relay		3	0
Communications Protocols Supported				3	0
Number of RTD Inputs (min.)				3	0

Notes
1. Fixed. Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data.
2. Vendor to Advise.
3. Preferred, Vendor to advise and provide reasoning for difference.
4. Equipment layout drawing shown is indicative. Vendor shall provide room layout complete with dimensions, including penetrations and subfloor cable ladder. Preliminary layouts to be provided with submission of tender.
5. Loading diagrams shall be provided with documentation, in accordance with note 6.
6. Loads at the loading points shall be provided as unfactored and for primary load cases (Self weight, Live load, Wind etc) and shall be in accordance with AS 1170. Designated loading points shall be specified by Vendor.
7. Power, control and earthing cables between equipment installed in the room to be run on cable ladder installed within the switchroom baseframe. Supply of this cabling and cable ladder to be by the Vendor.
8. Internal room lighting and small power cables to be run in duct/conduit internal to the room.
9. Communications block diagrams shall be provided post award. Refer to switchroom equipment list and cable schedule for indicative patch leads required.
10. HVAC sizing calculations to be performed by Vendor confirming size, selection and capacity of HVAC systems for given site conditions, equipment heat loads and passive heat gain.
11. HVAC design to confirm H_2 build up from battery charging (415VAC UPS and 48V DC battery) meets AS2676.2 ventilation requirements.

Table A Equipment Heat Load

No.	Switchroom	VSD Heat Load (kW) From Load List	Harmonic Filter Heat Load (kW)	Dry Transformer Heat Load (kW) per Enclosure	MCC & Misc. Equipment Heat Load (kW) *Vendor to Complete Blank Cells*	Environmental Heat Loads – Solar Radiation + Passive Heat Load (kW) *Vendor to Complete Blank Cells*	Total Building Heat Load (kW)
1	HV Switchroom	51.7	N/A	N/A	14.3		76
2	LV Switchroom	19.1	N/A	N/A	4	11.2	34.2
3	Transformer Room	N/A	N/A	14	2.8	3.2	20

Table B HVAC Cooling Capacity

No.	Switchroom	Unit Qty (Duty)	Unit Qty (Standby)	Sensible Cooling (kW) (Duty)	Sensible Cooling (kW) (Standby)	Total Sensible Cooling (kW)
1	HV Switchroom	7	1	81.2	11.6	92.8
2	LV Switchroom	3	1	34.8	11.6	46.4
3	Transformer Room	2	1	23.2	11.6	34.8

Table C Switchrooms Shipping Dimensions and Weights

No.	Switchroom	Length (mm)	Width (mm)	Overall Height (mm)	Internal Ceiling Height (mm)	Total Shipping Weight (kg)
1	F23-SR-601	~30000	5100	4200	3500	60000
2	Platform	~20000	2000	1800		10000

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart		1E	2	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected) including: • Switchroom, Stairs, Landings, Hand Rail Details • A/C Condensor Support Details • HVAC schematic • Welding Connections Details • Footing Details GA • Frame Work Elevations/Details • FIP Cause and Effect Table • VESDA and FIP GAs and Schematics • Fire System GA drawing showing the VESDA tubing and location of detectors, call points etc • Switchroom Floor Plan GA • Lifting Plan/Diagram • Base Frame GA • Cable Ladder GA		1E	10	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	10	1E
	B03	Cross-section		1E	8	1E
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)		1E	10	1E
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)		1E	10	1E
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams				
	D02	Schematic Drawings		1E	8	1E
	D03	Terminations / Loop Diagrams				
	D04	Panel Layout and Fabrication Details		1E	6	1E
	D05	Nameplate		1E	6	1E
	D06	Cable Block Drawings Fire System Zone Block Plan		1E	8	1E
	D07	Electrical and Instrumentation GAs including:		1E	6	1E
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule	1E			
	F02	Electrical Performance data sheet		1E	6	
	F03	Equipment List		1E	6	1E
	F04	Vendor's Surface Protection Details		1E	6	
	F05	Lubrication Data Schedule				
	F06	Material Schedule				

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information	1E			
	F12	Calculations				
	F13	Completed $_{Engineer}$ Datasheets	1E			
	F14	Motor Datasheets				
	F15	Electrical Equipment List				
	F16	Instrument List				
	F17	I/O List				
	F18	Communications Addressing and Signal Mapping List				
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans		1E	6	
	G02	Inspection and Test Procedures (generic)				
	G03	Quality Plans and Procedures (generic)		1E	4	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	4	
	G05	Manufacturer's Data Report (MDR)		1E	Delivery -4	
	G06	Material Certificates		In MDR		
	G07	Statutory Authority Certificates		In MDR		
	G08	Test Data and Other Certificates		In MDR		
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure		1E	Delivery - 4	
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List	1E	1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -4	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -4	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	Prelim 4 Delivery +1	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				